



Laboratory Exercise 03

From Neurons to Neural Networks: Biological Inspiration in AI

Course: ITAI 4374 — Neuroscience model for AI

Student: Duc Nguyen

Date: February 2, 2026

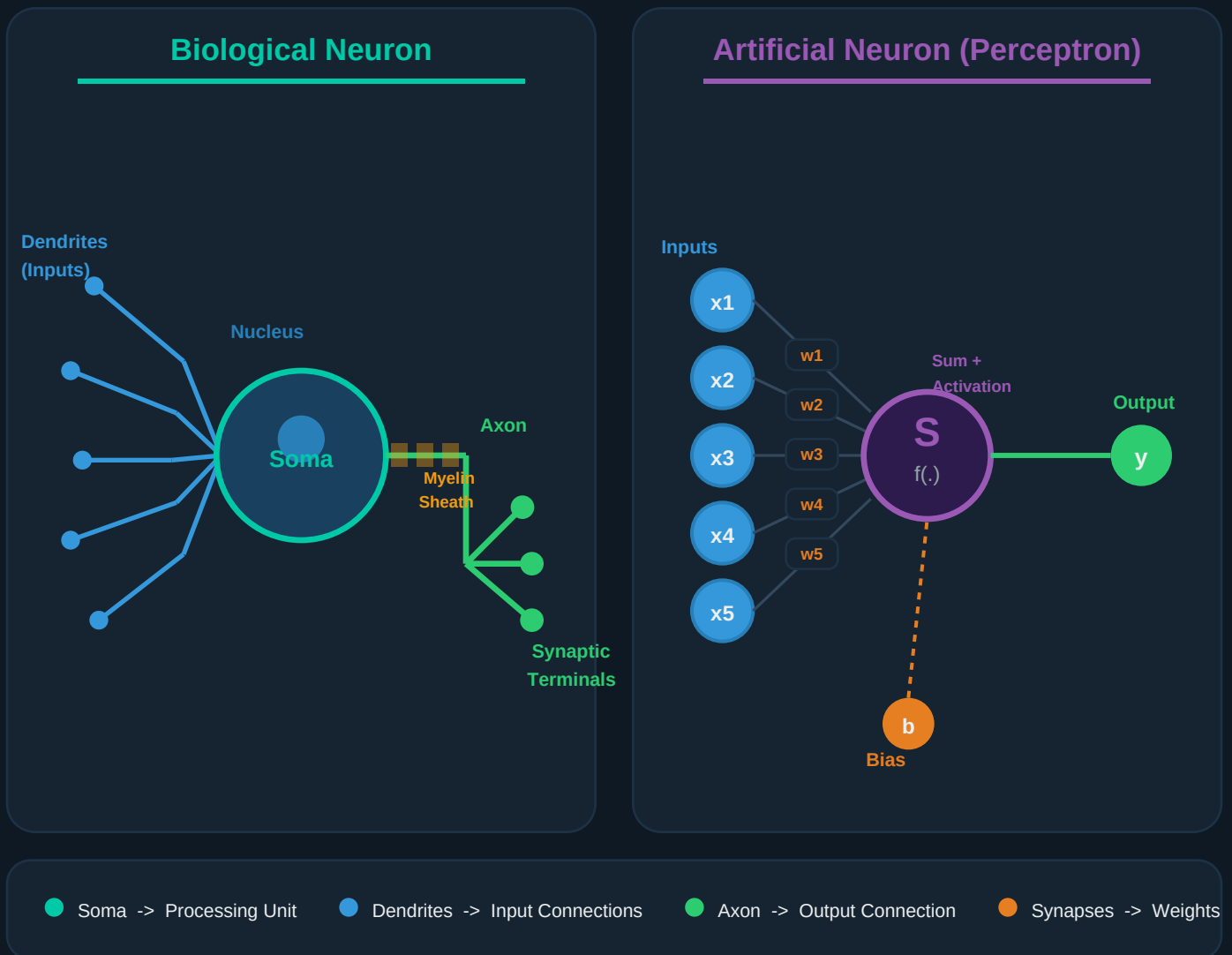
TABLE OF CONTENTS

1. Neural Network Comparison Analysis	3–5
1.1 Biological vs. Artificial Neuron Structure	3
1.2 150-Word Comparative Essay	4
1.3 Comparative Analysis Table	5
2. Brain Architecture and AI Design	6–8
2.1 Hierarchical Brain Organization	6
2.2 Regional Specialization & AI Parallels	7
2.3 Amygdala & Emotion Processing in AI	8
3. Integration Analysis & Future Implications	9–10
3.1 Improving AI Design via Biology	9
3.2 Critical System Comparison	9
3.3 Future Directions	10
R. References	11

1.1 Biological vs. Artificial Neuron Structure

Wolfram Alpha Reference: "neuron anatomy"

Figure 1: Side-by-side comparison of biological and artificial neuron architectures



1.2 Comparative Essay: Biological vs. Artificial Neurons

~150 words | Analysis of design inspiration, differences, and trade-offs

Comparative Analysis

Artificial neural networks draw direct inspiration from the biological neuron, yet the translation from biology to silicon is far from literal. The biological neuron integrates thousands of chemical signals across its dendrites, processes them electrochemically in the soma, and fires an all-or-nothing action potential along the axon — a dynamic, threshold-driven event. By contrast, the artificial neuron performs a static weighted sum of its numeric inputs and passes the result through a fixed activation function. This simplification was deliberate: it made mathematical analysis, gradient-based optimization, and hardware implementation tractable. However, it also discards rich temporal dynamics, metabolic self-repair, and the synaptic plasticity rules (such as spike-timing-dependent plasticity) that underpin biological learning. Biological neurons consume roughly $10^{(-15)}$ joules per spike — orders of magnitude less energy than a GPU performing the equivalent multiply-accumulate. Conversely, artificial networks achieve superhuman performance on narrow tasks through massive parallelism and data. The deepest AI advances will likely emerge from closing this gap: borrowing biological principles — energy efficiency, continual learning, and adaptability — while retaining the precision and scalability that artificial systems already offer.

● Biological Advantage

Energy-efficient ($\sim 10^{(-15)}$ J/spike),
adaptive, self-repairing

● Artificial Advantage

Scalable, precise, parallelizable,
mathematically tractable

● Shared Principle

Hierarchical feature extraction &
weighted signal integration

1.3 Comparative Analysis Table

Table 1: Biological Neural Networks vs. Artificial Neural Networks

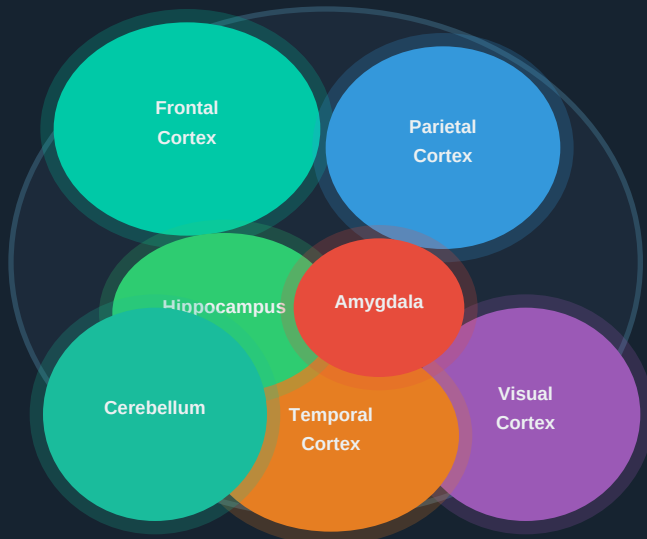
Characteristic	Biological Neural Network	Artificial Neural Network	AI Architecture Example
Signal Type	Electrochemical pulses (action potentials)	Continuous numeric values (floating-point)	All feedforward nets
Learning Rule	Hebbian / STDP ("neurons that fire together...")	Backpropagation & gradient descent	CNNs, Transformers
Processing Unit	Neuron soma integrates ~7,000 synaptic inputs	Perceptron: weighted sum + activation	Dense / FC layers
Connectivity	Sparse, dynamic (~1% connected)	Dense within layers, structured between	ResNet skip connections
Memory	Synaptic weight changes & structural plasticity	Parameter updates in weight matrices	LSTMs, Transformers
Energy Use	~10 ⁻¹⁵ J per spike (ultra-efficient)	~10 ⁴ – 10 ⁶ J per training run (GPU)	Neuromorphic chips (Intel)
Adaptability	Continuous lifelong learning; self-repair	Catastrophic forgetting; needs retraining	EWC, progressive nets
Temporal Dynamics	Rich spike-timing encodes information	Largely static per forward pass	Spiking Neural Networks
Generalization	Extremely robust; few-shot capable	Needs large data; transfer learning helps	Vision Transformers (ViT)

Table 1 — Compiled from Wolfram Alpha: "biological vs artificial neural network"

2.1 Hierarchical Brain Organization

Wolfram Alpha Reference: "human brain anatomy"

Brain Region Map



Level 1 — Sensory Input

Raw signals from eyes, ears, skin enter the thalamus and primary sensory cortices. -> AI parallel: Input layer / data pipeline

Level 2 — Feature Extraction

Visual cortex extracts edges, textures, shapes through hierarchical processing layers. -> AI parallel: CNN convolutional layers

Level 3 — Association & Memory

Temporal cortex & hippocampus integrate features and store learned associations. -> AI parallel: FC layers, LSTM memory cells

Level 4 — Decision & Action

Prefrontal cortex weighs options, plans actions; basal ganglia execute motor output. -> AI parallel: Output layer, RL policy

Figure 2 — Brain hierarchy & AI parallels | Wolfram Alpha "human brain anatomy"

2.2 Regional Specialization & AI Parallels

Mapping brain functions to modern deep-learning architectures

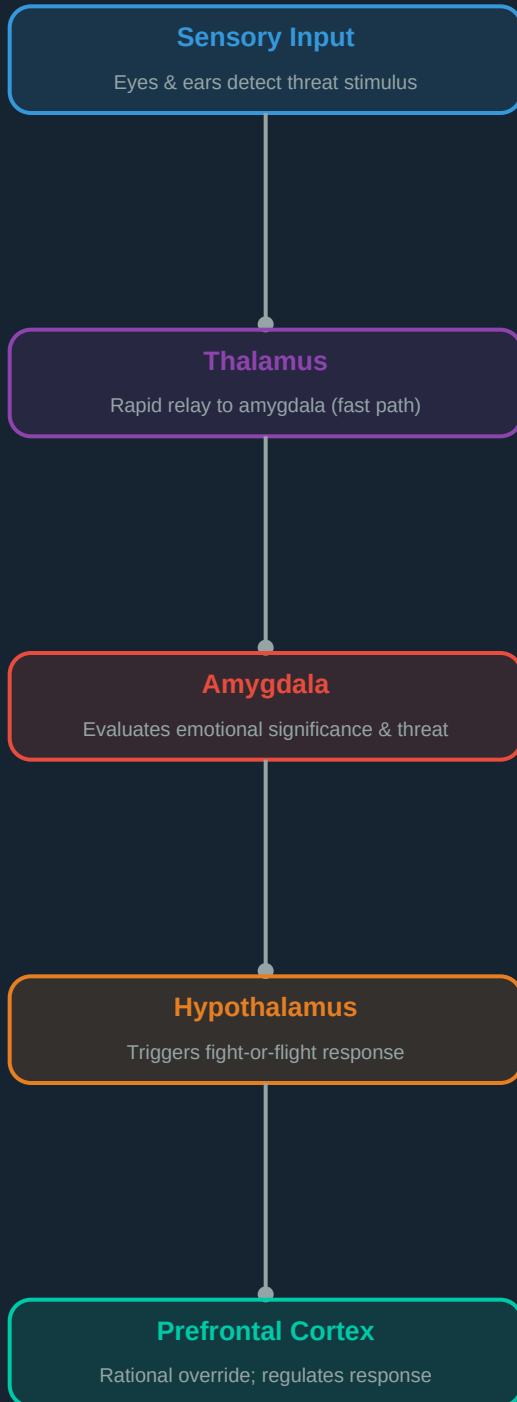


Figure 3 — Region-to-architecture mapping | Wolfram Alpha "visual cortex", "hippocampus", "prefrontal cortex"

2.3 Amygdala & Emotion Processing in AI

Wolfram Alpha Reference: "amygdala function" | How biological emotion informs AI

Amygdala Processing Flow



AI Decision Making

Emotion-aware AI can incorporate risk-weighted preferences, mimicking how the amygdala biases decisions toward survival-relevant outcomes.

Risk Assessment

Threat-detection modules inspired by rapid amygdala pathways enable real-time anomaly detection in security and autonomous systems.

Pattern Recognition

Emotional salience tagging (as in amygdala-cortical loops) helps AI prioritise high-value patterns in noisy, complex datasets.

Adaptive Learning

Amygdala-like reward/punishment signals can augment reinforcement learning, accelerating adaptation in dynamic environments.

Figure 4 — Amygdala signal flow & AI implications | Wolfram Alpha "amygdala function"

3 Integration Analysis & Future Implications

250-word synthesis | Biomimetic design, critical comparison, future directions

3.1 & 3.2 Improving AI Through Biology

Understanding biological neural networks offers a roadmap for solving some of AI's most persistent challenges. Current deep networks are remarkably powerful but fundamentally brittle: they require enormous datasets, consume vast energy, and forget previously learned tasks when trained on new ones — a phenomenon called catastrophic forgetting. The biological brain, by contrast, learns continuously, adapts with minimal data, and maintains knowledge across a lifetime. Biomimetic architectures — such as spiking neural networks (SNNs), neuromorphic hardware (e.g., Intel's Loihi), and meta-learning frameworks — attempt to transplant these principles into silicon. SNNs encode information in spike timing rather than continuous values, dramatically reducing energy consumption. Meta-learning algorithms ("learning to learn") mirror the brain's capacity for rapid few-shot adaptation. Neuroplasticity-inspired techniques like elastic weight consolidation (EWC) address catastrophic forgetting by protecting critical synaptic weights. Processing efficiency remains the starkest divide: a biological brain operates on ~20 W, while training a large language model can consume megawatt-hours. Generalization is another frontier — humans generalize from a handful of examples, while AI models often need millions. The future likely lies not in replicating the brain exactly, but in selectively borrowing its most powerful principles — energy-efficient coding, continual learning, and embodied sensorimotor integration — and combining them with the precision and scalability of conventional deep learning.

3.2 Critical Comparison



3.3 Future Directions

Emerging research frontiers at the intersection of neuroscience and AI

1

Neuromorphic Computing

Hardware that natively processes spikes — Intel Loihi, IBM TrueNorth — promises orders-of-magnitude energy savings over GPUs for edge AI.

2

Continual Learning

Algorithms that learn new tasks without forgetting old ones, mirroring the brain's lifelong plasticity. Key for robotics and personal AI.

3

Brain-Computer Interfaces

Bidirectional BCIs (e.g., Neuralink) could enable direct neural signal decoding, revolutionising prosthetics and human-AI collaboration.

4

Embodied AI

Robots that learn through physical interaction, similar to how biological organisms develop motor skills and spatial maps.

5

Affective Computing

AI systems that detect and simulate human emotions — informed by the amygdala-cortical circuit — enabling empathetic human-AI interaction.

6

Sparse & Efficient Models

Biologically inspired sparsity (~1-5 % of neurons active at once) applied to AI yields leaner models with lower latency and memory footprint.

The convergence of neuroscience and AI is accelerating. By bridging the gap between biological elegance and computational power, the next generation of AI systems will be more energy-efficient, adaptive, and aligned with the complexity of the real world.

References

Wolfram Alpha Queries

- 1 Wolfram Alpha. (2026). Neuron anatomy. <https://www.wolframalpha.com>
- 2 Wolfram Alpha. (2026). Artificial neuron perceptron. <https://www.wolframalpha.com>
- 3 Wolfram Alpha. (2026). Human brain anatomy. <https://www.wolframalpha.com>
- 4 Wolfram Alpha. (2026). Visual cortex. <https://www.wolframalpha.com>
- 5 Wolfram Alpha. (2026). Hippocampus. <https://www.wolframalpha.com>
- 6 Wolfram Alpha. (2026). Prefrontal cortex. <https://www.wolframalpha.com>
- 7 Wolfram Alpha. (2026). Amygdala function. <https://www.wolframalpha.com>
- 8 Wolfram Alpha. (2026). Biological vs. artificial neural networks. <https://www.wolframalpha.com>

Textbook & Peer-Reviewed Academic Sources

- 1 Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep learning. MIT Press.
- 2 Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- 3 Kandel, E. R., Koester, J. D., Mack, S. H., & Siegelbaum, S. A. (2021). Principles of neural science (6th ed.). McGraw-Hill.
- 4 Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., ... Hadsell, R. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 3521–3526. <https://doi.org/10.1073/pnas.1611835114>
- 5 LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444. <https://doi.org/10.1038/nature14539>
- 6 Maass, W. (1997). Networks of spiking neurons: The third generation of neural network models. *Neural Networks*, 10(9), 1659–1671. [https://doi.org/10.1016/S0893-6080\(97\)00011-7](https://doi.org/10.1016/S0893-6080(97)00011-7)
- 7 Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, ■., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems (NeurIPS)*, 30, 5998–6008.

Industry & Emerging Technology Sources

- 1 Davies, M., Srinivasa, N., Lin, T. H., Chinya, G., Cao, Y., Choday, S. H., ... Wang, H. (2018). Loihi: A neuromorphic manycore processor with on-chip learning. *IEEE Micro*, 38(1), 82–99. <https://doi.org/10.1109/MM.2018.112130359>